

Software Engineering

Lab 1 - Requirements Engineering and UML Use-Case Modelling

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Problem Statement : #15 - Pharmacy Expiry & Re-order Dispatch Engine
Repository : <https://github.com/RohanMay-Suresh/SE-Labs-PES1UG24CS383>

Problem Statement

A pharmacy stock management engine that tracks batch expiry dates, generates First-Expired-First-Out (FEFO) dispensing lists, and automatically raises purchase orders when stock falls below a threshold. Actors: Pharmacy Clerk, Inventory Supplier, and a Scheduler that runs the daily expiry sweep.

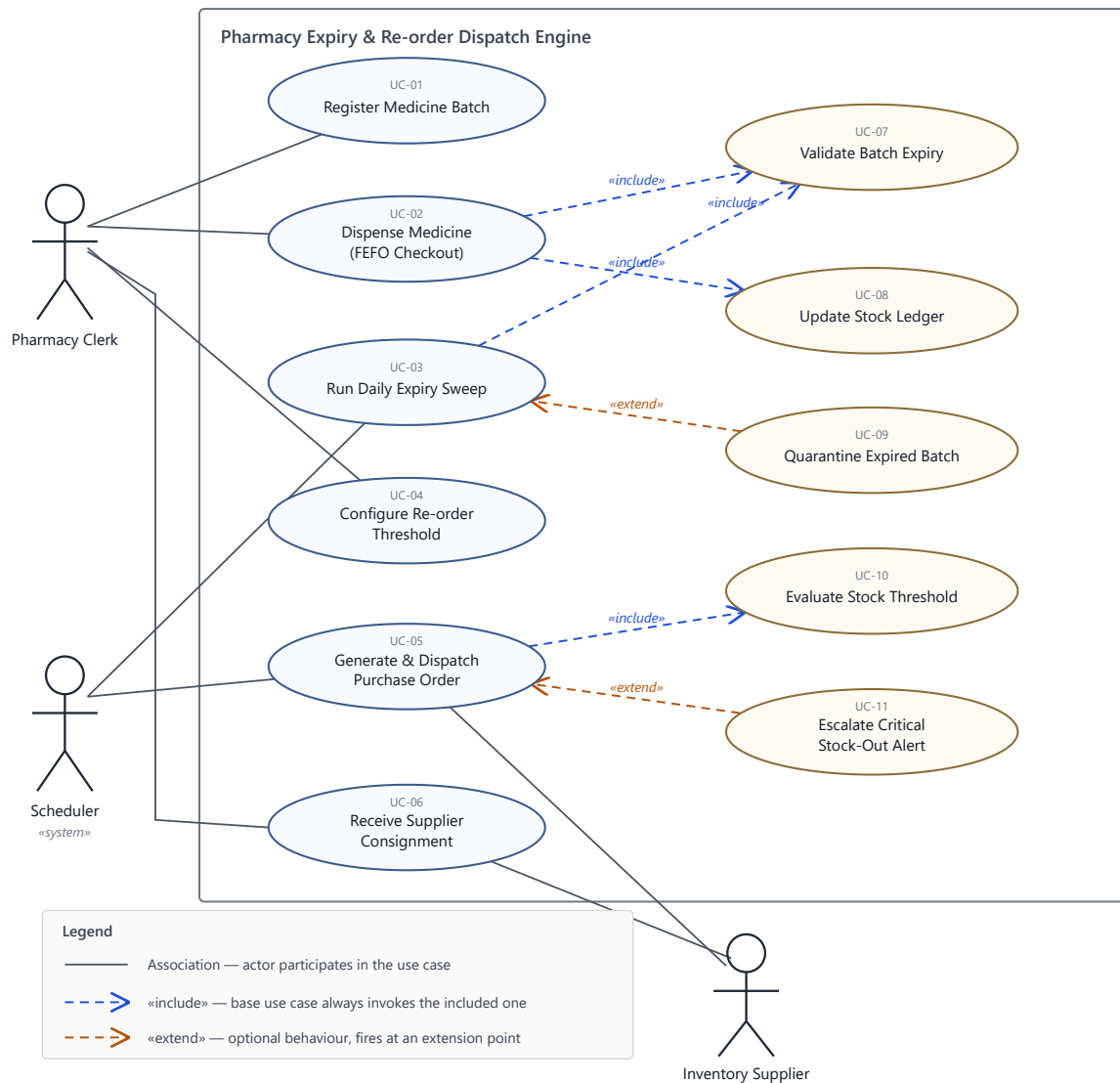
1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria (measurable pass/fail)	Rationale
FR-001	Functional	The system shall enforce FEFO picking order by prioritizing medicine batches with the nearest expiry date during sales checkout.	High	Pass: For every dispense line, the oldest valid (unexpired, non-quarantined) batch of that medicine is selected first. Fail: A newer batch is assigned while an older unexpired batch of the same medicine still holds stock.	First-Expired-First-Out is the standard hospital stock-rotation rule. Manual picking lets near-expiry stock age on the shelf until it must be written off, so the ordering must be enforced by the system rather than left to the clerk's judgement.
FR-002	Functional	The system shall capture batch number, manufacturer, quantity and expiry date for every medicine batch registered into stock, and shall reject any batch whose expiry date is on or before the date of registration.	High	Pass: Batch is stored with all four attributes plus a system-generated batch ID and status AVAILABLE; a batch submitted with a missing attribute or a past / same-day expiry date is rejected with a validation message and is not persisted. Fail: A batch is accepted with a null expiry date, or an already-expired batch enters AVAILABLE stock.	FR-001 and FR-003 are only as reliable as the data captured at intake. A batch with no expiry date is invisible to FEFO ordering, and admitting already-expired stock creates a direct patient-safety hazard.
FR-003	Functional	The system shall run a daily expiry sweep that flags every batch expiring within the configured near-expiry window (default 90 days) as NEAR_EXPIRY, and shall move every batch past its expiry date to QUARANTINED, making quarantined batches unavailable for dispensing.	High	Pass: After the sweep, every batch whose expiry date has passed holds status QUARANTINED and is excluded from FEFO selection; every batch expiring within the window appears on the near-expiry alert list. Fail: An expired batch remains dispensable, or a batch inside the window is absent from the alert list.	Expiry is a date-driven event with no user action behind it, so detection must be automatic. Quarantining rather than deleting preserves the batch for supplier return and for the audit trail required by NFR-002.
FR-004	Functional	The system shall allow the Pharmacy Clerk to configure a safety-stock threshold and a re-order quantity per medicine item, and shall re-evaluate dispensable	Medium	Pass: Threshold and re-order quantity are persisted per item; a re-order trigger is raised within the same transaction as the movement that takes dispensable stock below the	Consumption rates differ per drug, so a single global threshold would over-stock some items and starve others. Excluding quarantined stock is essential — otherwise expired

		stock against that threshold after every stock movement.		threshold. Fail: The threshold is not applied, or the evaluation counts QUARANTINED / expired batches as available stock.	batches mask a real shortage and the re-order in NFR-001 never fires.
FR-005	Functional	The system shall track each dispatched purchase order through the states DISPATCHED, ACKNOWLEDGED, RECEIVED and CLOSED, and on receipt shall reconcile the delivered quantity against the ordered quantity, raising a short-delivery discrepancy record when they differ.	Medium	Pass: A purchase order reaches CLOSED only after the delivered batches have been registered via FR-002 and the quantities match; a mismatch produces a discrepancy record carrying the variance and leaves the order open. Fail: A purchase order is auto-closed on dispatch, or a quantity mismatch is accepted silently.	Closes the loop opened by NFR-001 — without receipt tracking the engine cannot tell an order in transit from an order never fulfilled, and would re-order the same item repeatedly. The discrepancy record also gives the pharmacy evidence for supplier billing disputes.
NFR-001	Nonfunctional (Performance & Security)	The system shall automatically generate and dispatch supplier purchase order requests when item stock dips below safety threshold.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load — the purchase order is generated and transmitted within 60 seconds of the FR-004 trigger while sustaining 500 concurrent stock movements, over TLS 1.2+ with a digitally signed payload. Fail: Dispatch latency exceeds 60 s at target load, or a purchase order is transmitted unsigned or over an unencrypted channel.	Re-ordering is only useful if it beats the supplier's daily cut-off; latency translates directly into stock-out days. A purchase order is a financially binding instruction to an external party, so transport encryption and payload signing are needed to prevent interception or tampering.
NFR-002	Nonfunctional (Reliability & Auditability)	The system shall record every stock movement (intake, dispense, quarantine, purchase-order dispatch, receipt) as an immutable, timestamped, user-attributed audit entry retained for 5 years, and shall remain available for at least 99.5% of pharmacy operating hours (07:00–22:00).	High	Pass: A 1000-transaction regression run produces one audit entry per movement carrying actor ID and UTC timestamp; a direct UPDATE or DELETE against the audit store is rejected; measured monthly downtime inside the operating window is 4.5 hours or less. Fail: Any movement completes without an audit entry, or an existing audit row can be altered.	Controlled-drug regulation and hospital accreditation require a reconstructible history of who moved which batch and when; a mutable log would not survive an audit. The availability target reflects that dispensing halts entirely when the engine is down, since FEFO order cannot be determined manually at counter speed.

2. UML Use-Case Diagram

11 use cases across 3 actors. UC-02 and UC-03 both include UC-07 (Validate Batch Expiry); UC-02 also includes UC-08 (Update Stock Ledger) and UC-05 includes UC-10 (Evaluate Stock Threshold). UC-09 extends UC-03 only when an expired batch is found, and UC-11 extends UC-05 only when stock has already hit zero.



3. Use-Case Flow Specification

The flow below is for UC-02, Dispense Medicine (FEFO Checkout), which realises FR-001.

Use-Case ID	UC-02
Use-Case Name	Dispense Medicine (FEFO Checkout)
Primary Actor	Pharmacy Clerk
Trigger	The Pharmacy Clerk submits a prescription or counter sale for checkout.
Related Requirements	FR-001 (FEFO ordering), FR-004 (threshold re-evaluation), NFR-002 (audit trail)

Preconditions

1. The Pharmacy Clerk is authenticated and holds the DISPENSE permission.
2. A valid prescription or counter-sale record exists, listing medicine codes and quantities.
3. At least one batch of each requested medicine exists with status AVAILABLE.
4. The daily expiry sweep (UC-03) has completed for the current date, so batch statuses are current.

Postconditions

1. The requested quantity is deducted from the nearest-expiry AVAILABLE batch(es) of each medicine.
2. The stock ledger is updated and an immutable audit entry is written for every deduction, attributed to the clerk and timestamped (NFR-002).
3. Dispensable stock for each affected item is re-evaluated against its safety threshold; a re-order trigger is raised where the threshold has been breached (FR-004).
4. A dispense slip is generated listing, per line, the medicine, quantity, batch ID and expiry date issued.

On failure: No stock is deducted, no ledger or audit entry is written, and the sale remains open — a dispense transaction is all-or-nothing across every line in the sale.

Main Success Scenario

Step	Actor	Action
1	Clerk	Opens the checkout screen and selects the prescription or sale record to dispense.
2	System	Displays each requested medicine with the quantity to be dispensed.
3	Clerk	Confirms the dispense request.
4	System	For each line, retrieves all batches of that medicine with status AVAILABLE and sorts them by expiry date ascending, breaking ties by earliest registration date.
5	System	«include» UC-07 Validate Batch Expiry — re-checks the selected batch against the current system date to confirm it has not expired since the last sweep.
6	System	Allocates the requested quantity to the nearest-expiry valid batch and displays the proposed allocation (medicine, batch ID, expiry date, quantity) for confirmation.
7	Clerk	Confirms the allocation.

8	System	«include» UC-08 Update Stock Ledger — deducts the allocated quantity, writes the ledger movement and the audit entry, and commits the transaction.
9	System	«include» UC-10 Evaluate Stock Threshold — compares remaining dispensable stock against the configured safety threshold and raises a re-order trigger if it has been breached.
10	System	Generates and prints the dispense slip, and returns the clerk to the checkout screen.

Alternate Flow - 6a. Nearest-expiry batch holds insufficient quantity

Step	Actor	Action
6a.1	System	Detects that the nearest-expiry valid batch holds less than the requested quantity.
6a.2	System	Allocates the full remaining quantity of that batch, then moves to the next batch in ascending expiry order and repeats until the requested quantity is fully covered, preserving FEFO order across the split.
6a.3	System	Displays the multi-batch allocation, showing the quantity drawn from each batch with its batch ID and expiry date.
6a.4	Clerk	Confirms the split allocation. The flow rejoins the main success scenario at step 8, where all deductions in the split are committed as a single atomic transaction.
6a.5	System	If the combined stock across all valid batches is still short, dispenses nothing, informs the clerk of the maximum dispensable quantity, and raises a re-order trigger for the item (FR-004).